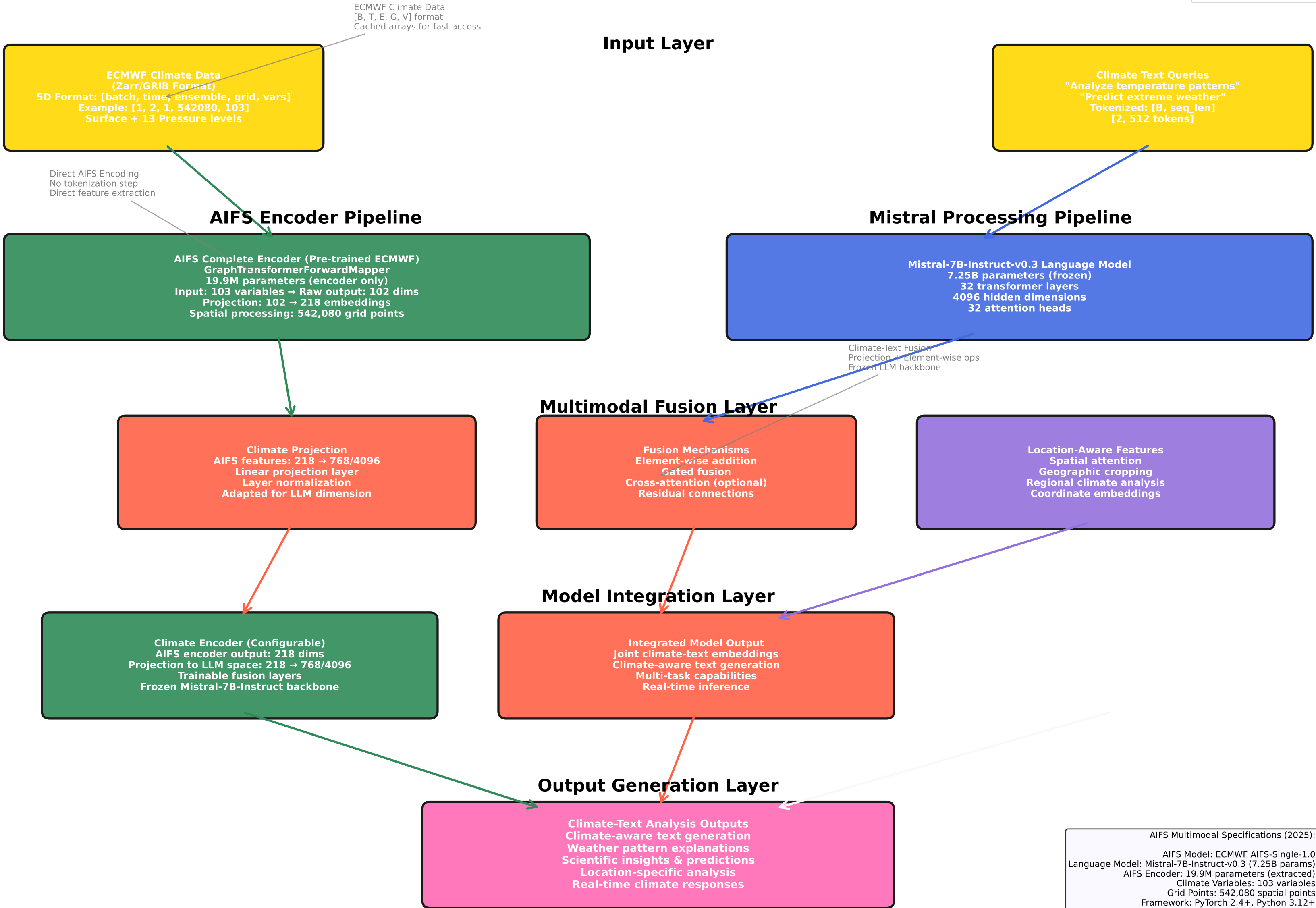


AIFS Multimodal Climate AI Architecture (2025)

ECMWF AIFS Encoder + Mistral-7B-Instruct-v0.3 Integration

- AIFS Components
- Mistral Components
- Encoder/Processing
- Fusion Mechanisms
- Data Sources
- Outputs



AIFS Multimodal Specifications (2025):

AIFS Model: ECMWF AIFS-Single-1.0
Language Model: Mistral-7B-Instruct-v0.3 (7.25B params)
AIFS Encoder: 19.9M parameters (extracted)
Climate Variables: 103 variables
Grid Points: 542,080 spatial points
Framework: PyTorch 2.4+, Python 3.12+
Training: CPU optimized, GPU compatible